Reg Environ Change (2013) 13:537–550
DOI 10.1007/s10113-012-0343-0

ORIGINAL ARTICLE

Rural organizations and adaptation to climate change and variability in rural Kenya

Camille Washington-Ottombre · Bryan C. Pijanowski

Received: 27 May 2011 / Accepted: 25 August 2012 / Published online: 17 October 2012
© Springer-Verlag 2012

Abstract Numerous studies have shown that collective action affects the type and efficiency of short- and long-term adaptation to climate change. This empirical study contributes to the body of the literature on collective action and adaptive capacity by demonstrating how organizations frame responses to climate variability and change in rural Kenya by promoting local rural institutions. By analyzing interviews, role-playing games, and household surveys, we ask how local rural organizations shape coping strategies to climate variability and how they may structure future adaptations to climate change. We also investigate what types of households participate in those organizations and how their participation may impact their vulnerability to climate change and variability. Our analysis shows that in places rendered especially vulnerable to climate change by arid climatic conditions, the disengagement of governmental services, and a limited access to income-generating activities, local rural organizations increase livelihood security. Those organizations reduce local vulnerabilities and enhance collective action. In contrast to common diversification and livelihood security strategies which rely on the access to urban or peri-urban structures, local rural institutions and organizations allow for rural and grassroots sustainable adaptation strategies. In that respect, they constitute a resilient and mostly untapped resource for visibly strengthening livelihood security and adaptive capacities in rural Kenya.

C. Washington-Ottombre (✉)
Environmental Studies Program, Oberlin College,
Oberlin, OH 44074, USA
e-mail: cwashing@oberlin.edu

B. C. Pijanowski
Department of Forestry and Natural Resources,
Purdue University, West Lafayette, IN 47906, USA

Keywords Collective action · Adaptation to climate change · Local rural institutions · Rural organizations · Rural producer organizations · Kenya

Introduction

Future climate change will challenge societies around the world. In particular, numerous researchers and policy experts have suggested that people in Sub-Saharan Africa (SSA) whose livelihoods are tied to climate-dependent ecosystem services are highly vulnerable (Osman Elasha 2006; Easterling et al. 2007; Schneider et al. 2007; Adger et al. 2006; Brown and Funk 2008). Equally important are the global, regional, and local institutions that affect those communities' vulnerability and adaptive capacity with regard to climate change (Adger 1999; Adger et al. 2005). Therefore, a better understanding of the institutions, defined by Ostrom (2005, 2010) as the rules used by individuals to order their relationships and refer to the actions that are required, prohibited, or permitted in a specific context, is crucial to understanding the adaptive capacity of people in SSA.

Institutions shape collective action and in turn influence adaptive capacity and adaptation to climate change (Adger et al. 2005). Although empirical research on how collective action is instrumentalized to adapt to climate change is limited, a significant body of research has explored household coping and adaptation strategies and can be used to frame explorations of the role of institutions in shaping adaptive capacity (Corbett 1988; Chambers and Conway 1992; Smithers and Smit 1997; Adams et al. 1998; Campbell 1999; Adger 1999; Kelly and Adger 2000; Eriksen et al. 2005). Along with the role of physical capital, the past literature on coping strategies and adaptation

to climate variability and change has emphasized the importance of social capital and the role of institutions in shaping household responses to shocks and stressors. Institutions are part of the entitlements which determine household vulnerability to perturbations (Adger 1999). Even if as short-term coping strategies, households rely mostly on physical and social capital (Corbett 1988; Adams et al. 1998), the evolution of coping strategies as well as long-term adaptations are molded by institutional settings and efficiency (Smithers and Smit 1997; Campbell 1999; Adger 1999; Kelly and Adger 2000; Eriksen et al. 2005).

This paper applies Agrawal's (2008) "Climate Adaptation, Landscape, and Institutions" framework to adaptation strategies at the household level for people located in communities along a climatic gradient on Mt. Kenya (Kenya). According to Agrawal, local institutions shape adaptation to climate change in various landscapes in three different ways. First, they structure the impacts of climate change and the vulnerability of a human group. Second, they promote adaptation practices related to the common pooling of resources and their mobility, storage, diversification, and exchanges. More specifically, local institutions accomplish this by creating an incentive framework for individual and collective action and mediate individual and collective responses to climate change. Third, they provide access to or deliver external resources for adapting to climate change. Institutional linkages in the local institution itself, with other local institutions, and with institutions at higher levels, are crucial to adapting to climate change because they allow for diversity in the flow of resources toward vulnerable groups.

Agrawal (2008) emphasizes the operational role of Local Rural Institutions (LRIs) in shaping adaptation to climate change within a polycentric climate change governance order. He identifies the need for more empirical studies on the type of households that participate in those institutions in order to determine who will be more vulnerable to climate change. In order to address those needs, this study analyzes the role of LRIs in shaping adaptive capacities in rural areas of Kenya and identifies the characteristics of households which participate in those institutions. More specifically, we ask how LRIs shape coping strategies to climate variability and how they may structure future adaptations to climate change. We also examine what types of households participate in organizations that promote those institutions and ask what impact their participation may have on their vulnerability to climate change and variability. In order to address those questions, we first analyze interviews of farmers that explore social participation and coping strategies as well as role-playing games that simulate future adaptation to climate change. We then examine a survey on coping strategies to investigate how households' entitlements and access to information about farming and other income-generating activities from various institutional sources determine their participation in rural organizations that promote LRIs.

LRIs and RPOs in Kenya

Local rural institutions (LRIs) are supported by organizations characterized by a certain scale of person-to-person interaction as well as a specific geographic and economic frame of reference. Those organizations are situated above the individual and household level but below the sub-district, district, provincial, national, and international levels. As Uphoff and Buck (2006) note, organizations that promote LRIs allow for face-to-face interactions between individuals. In terms of geographic and economic scale, rural areas differ by definition from non-urban areas and are characterized by low population density and economic activities that are traditionally organized around primary production often plagued by market imperfections. Those organizations exist in the public, private, and civic sector, in the form of membership organizations or cooperatives. In recent years, a new hybrid category called rural producer organizations (RPOs) has appeared. RPOs are characterized by the diversity, complexity, and uniqueness of their genesis, functions, organizational structures, and path of evolution (Bosc et al. 2002). In short, they do not reproduce a pre-established organizational framework—such as the cooperative model—but display and operationalize local specificities. Hence, RPOs promote LRIs, are situated across the public, private, and civic sector, and are characterized by highly local, specific, and evolving structures and functions.

RPOs have become more prominent in the recent past in developing countries as a response to both governmental disengagement and a more uncertain and global environment (Bosc et al. 2002). In that perspective, RPOs can be seen as filling the gap left by failing agricultural extension services (Moris 1991). The growth of RPOs have accompanied the diversification of rural activities toward the integration to a global economy and away from a primary production framed by government agencies. RPOs have the necessary local "fine-tuning" to produce a variety of needed public goods (Uphoff and Buck 2006). In rural areas within developing countries, RPOs directly provide information, resources, services, and opportunities to rural households and often facilitate access to public goods. They can create a favorable environment for a sustainable development and adaptation to perturbations such as climate change.

While Kenya has a long history of rural group activities, organizations such as RPOs constitute an entirely new type

of local collective action. For the first time in Kenya's post-colonial history, RPOs constitute a collective effort and grassroots organization that structures local livelihood security. In colonial times, local populations started to perceive collective action negatively because they associated the concept with the collective forced labor that the British used to dig roads and terraces in rural zones like the Mt. Kenya area (Mwaniki 1975). After Kenyan independence, Jomo Kenyatta promoted a new institution called *Harambee* which literally means "all put together" in Kiswahili. The objective of the Harambee movement was to generate local development projects but it often lacked adequacy and free participation from local populations (Haugerud 1993).

In contrast, RPOs truly are grassroot organizations that result in local collective action. Life story interviews conducted by Washington-Ottombre (2010) suggest that RPOs began to appear on Mt. Kenya as grassroot organizations after "the emergency," the period of colonization during which households in the Mt. Kenya area were confined by the British in detention camps for several years. At this time, women decided to form small saving and labor groups to rebuild the villages, and this mutual help system, albeit under a different form, continued to operate after Kenyan independence. The action of those women's groups remained punctual and restricted to short-term hazard coping strategies. Since the mid-1980s, however, RPOs began to grow, moving forward both the Harambee movement and the model of women's groups.

Around Mt. Kenya, rural organizations consist mostly of cooperatives, church groups, family groups, and RPOs. All rural organizations, except for those cooperatives that are structured by national government agencies, have a written constitution that has to be approved and hosted by a supervising organization such as the Ministry of Social Affairs, a church office, or an NGO. The constitution describes the scope of action and operation of the rural organization, membership conditions, as well as the sanctions imposed on contravening members. All rural organizations are organized similarly, with a chairperson and a vice-chairperson occupying leadership positions and with a treasurer and a secretary assisting those leaders. Group meetings are generally organized weekly, although many hold frequent meetings when emergencies arise.

RPOs are distinct from other rural organizations as they are characterized by a wide diversity of local arrangements that evolve over time, adapting to new circumstances. Three categories of RPOs exist in rural Kenya. First, rotating savings and credit organizations (ROSCAs), or "merry-go-rounds" as Kenyan farmers call them, are groups that meet regularly to collect and redistribute funds equally among members who take turns, one member recipient per meeting, receiving the decided sum. (see for instance Besley et al. 1993; Bouman 1995; Mimuyu 1999; Gugerty 2007). A second category of RPOs includes saving groups that act as micro-credit organizations in which every member contributes to the common account and can be granted a loan secured by personal assets and repaid with interest (Armendáriz and Morduch 2007). The analysis of our survey on social participation (Washington-Ottombre 2010) shows that saving groups are more formally organized than ROSCAs; rely less on neighborhood, family, or kin ties; tend to circulate higher sums of money; and operate with the formal banking system. Finally, self-help groups occupy a third category of RPOs. These aim to pool resources to produce income-generating activities (Mwaniki 1975). Our survey indicates that, like saving groups, self-help groups are formally structured and do not rely on neighborhood, family, or kin ties. Most self-help groups, from their very creation, integrate the functions of ROSCAs and saving groups in order to maintain the short-term motivation of the members and slowly build income-generating activities. In addition, self-help groups are characterized by their high ability to evolve beyond their initial purpose and encompass an increasing number of activities.

Study area description

This study was conducted in three villages on the Eastern slopes of Mt. Kenya (Kenya) situated along a climatic gradient from humid to semi-arid (Fig. 1) (Olson 2003). The first, Ndunduri, is located in a humid area near the top of Mt. Kenya. In those highlands, farmers have been cultivating coffee and tea as cash crops for more than a generation. The village is the richest of the three study sites but suffers from land scarcity and the competition between subsistence and cash crops. The second, Kambita, is located at the foot of Mt. Kenya in a semi-arid area. The recent introduction of miraa (*Catha edulis*) crops, a legal drug cultivated for export to the Arabian Peninsula, has brought relative prosperity to the village along with numerous social problems (Maitima et al. 2004; Carrier 2005; Beckerleg 2006). The third, Gatumbiri, is also located in a semi-arid area. Gatumbiri farmers cannot cultivate miraa due to the presence of sandy soils and do not rely on any cash crop for income. The village is characterized by severe poverty, malnutrition, and high male out-migration rates. During the 2008–2009 study, all villages suffered from a two-year drought which resulted in little or no harvest of food crops. In addition, all villages had suffered from numerous land evictions, administrative re-districting, and the economic crisis that resulted from the tumultuous 2007 presidential elections (Amnesty International 2007).

These study sites are particularly relevant for the study of adaptation to climate change and variability for three

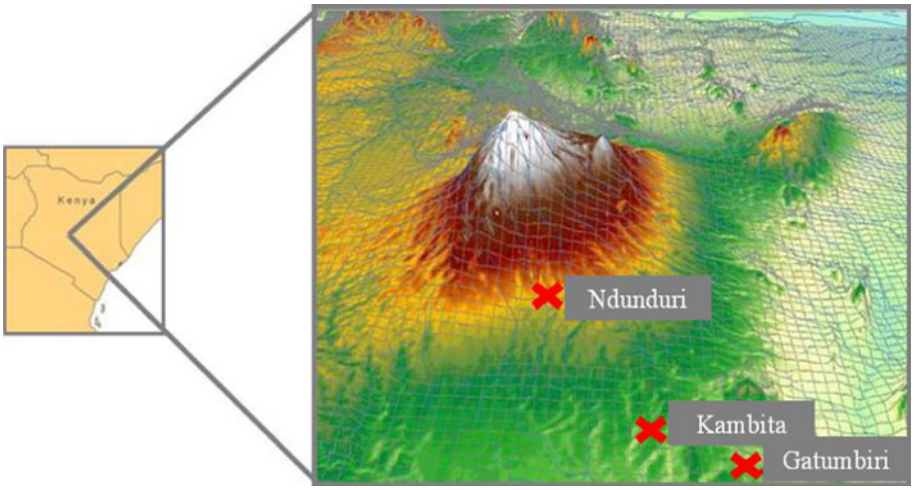


Fig. 1 Location of the villages along the slopes of Mt. Kenya, Kenya

reasons. First, they are located on a climatic gradient from humid to semi-arid which allows for exploring the impact of various forms of physical capital in adaptation to climate change (Maitima et al. 2004). Second, each site has experienced highly dissimilar paths of development, displaying a wide variety of distinct social and economic characteristics (Olson et al. 2003). Third, climate projections suggest that highland communities, such as those located along the Eastern slopes of Mt. Kenya, are likely to experience significant climate change (Olson et al. 2008; Moore et al. 2010; Moore et al. in press).

Methods

Semi-directed interviews and role-playing games

Semi-directed interviews (Holstein and Gubrium 2003) of farmers were conducted in 2008–2009 to investigate paths for adaptation to climate change and variability in the study area. Ten to 15 farmers and local leaders were interviewed in English in each community. We invited interviewees to identify organizations that, in their opinion, had shaped or will shape past, present, and future adaptations to climate change and variability. Several LRIs and RPOs were identified (Washington-Ottombre 2010).

Role-playing games (RPGs) were used as a participatory technique for simulating adaptation to climate change (Berkes and Jolly 2001; Washington-Ottombre 2010; Washington-Ottombre et al. 2010). Employed as a simulation tool in social sciences, RPGs can be defined as "the performance of an imaginary or realistic situation played by people with given roles in order to analyze behavioral patterns" (Shaftel and Shaftel 1967). In these games, farmers were confronted with various climatic scenarios and had to mobilize their social environment to adapt their land-use practices and income-generating activities to climate variability and change. A series of RPGs were conducted in Ndunduri and Kambita villages in order to observe various paths of adaptation in these two distinct climatic zones, humid (Ndunduri) and semi-arid (Kambita). Each game simulated 10–12 seasons during which male and female farmers had to make land-use and income-generating decisions within their social environment. The simulations followed a climatic scenario of bimodal rains for four seasons followed by a prolonged drought of six to eight seasons. The game (see Washington-Ottombre 2010 for more details) was followed by individual and collective debriefings to explore the decision-making process and to compare and contrast the dynamics that occurred in the game with real-life situations.

Household surveys

Household surveys were conducted in the local dialect through directed interviews in the three villages: Gatumbiri (125 respondents), Kambita (77 respondents), and Ndunduri (197 respondents). The sample size was estimated to equal 90–100 % of each village. Information was collected on household participation in rural organizations along with demographics, assets and vulnerability, access to information related to farming and other income-generating activities, and other forms of social participation.

Our dependent variable, participation in rural organizations, measures the involvement of households in a range of rural organizations. The dependent variable is coded as follows: 1 stands for no participation in rural organization, 2 stands for participation only in church groups, 3 stands for participation in church groups and ROSCAs, 4 stands for participation in church groups, ROSCAs, and cooperatives, and 5 stands for participation in church groups, ROSCAs, and RPOs. In order to test our model and elucidate participation of different types of households in different kinds of rural organizations, we placed our

Fig. 2 Statistical model

$$Participation = \alpha + \beta \cdot Demographics + \delta \cdot Household's\ Vulnerability + \gamma \cdot Access\ to\ information + \varepsilon,\quad \varepsilon \sim N(0, \sigma^2),$$

Where:

Participation= Degree of access to local rural institutions and rural producer organizations
Demographics= (Gender, Age, Education)
Household's vulnerability= (Size of land, Main occupation of head of household, Main location of head of household, Part of income from farming, Number of people giving remittances to the household)
Access to information= Frequency of access to information related to farming and other income generating activities from various organizations and institutions

independent variables in 3 categories: demographics, household entitlements, and access to information relative to farming and other income-generating activities (Fig. 2). First, social and human capital, as they relate to vulnerability, is reflected in demographics such as gender, age, and education. The past literature emphasizes the fact that women, elderly, young people, and people with no education are more vulnerable to climate change (Kelly and Adger 2000; Adger 2006). Second, the category of variables titled "household entitlements" reflects other factors that impact household adaptation to climate change and variability. Our survey attempted to determine the role of the main occupation of household heads as well as the share of farming activities in household income in order to assess the adaptation path of resource-dependent groups (Leary 2006; Adger et al. 2006). Similarly, the role of the primary location of the occupation of household heads as well as the reception of remittances was analyzed in order to assess the role of migration in livelihood security (Adger 1999). Finally, the third category of variables titled "access to information relative to farming and other income-generating activities" follows past studies on LRIs that have examined how the growth of those institutions was in part a direct response to the institutional vacuum faced by rural households (Bosc et al. 2002). Therefore, a third group of variables explored the frequency of information about farming and other income-generating activities received by farmers from various institutional sources.

A Classification and Regression Tree (CART) model was developed for each community. CART is a nonparametric technique that can select from among a large number of independent variables those that are most important in determining the dependent variable, and their interactions. CART models (De'ath and Fabricius 2000) are simple rule-based statistical models that split data within a particular independent variable into two homogenous groups depending on how they interact with the dependent variable and arrange them hierarchically. Many variables can be split and arranged hierarchically. Dependent variables can be categorical (classification) or continuous (regression). Independent variables can be of any type (binary, ordinal, categorical, or continuous), and no assumptions about the structure of the data are needed. Several rules are generally applied to determine tree size; these include the proportion in reduction of error (PRE) that is needed to create a split, maximum tree size and minimum sample size of a terminal node. Trees are represented as a graphic, with means, standard error of the mean, and sample size of each node being generally reported. We used a least squares loss model, setting the proportion reduction of error at 0.02. Systat 13 was used.

Results

Semi-directed interviews

Interviews emphasize the prevalent role of RPOs in crafting coping strategies and adaptation (see Washington-Ottombre 2010 for more details). Our analysis highlights the fact that RPOs in the study area focus on the five adaptation practices described by the literature (Agrawal 2008): diversification, communal pooling of resources, exchange, storage, and mobility (Table 1). Most RPOs focus on the diversification of income-generating activities in order to reduce local vulnerabilities. In a context of widespread soil erosion, numerous RPOs maintain tree nurseries for instance. Others have increased the market value of livestock by adopting new dairy breeds; the sale of which constitutes a traditional coping strategy. Finally, a few have experimented with new cash crops such as watermelons, passion fruit, grafted mangoes, or papaya.

Through the communal pooling of resources, RPOs also undertake other endeavors that in turn promote the diversification of income-generating activities. They offer saving or ROSCAs activities in order to develop income diversification strategies. These can act as the main function of the RPO, or they can serve to sustain other

Table 1 The role of LRIs in shaping past and present coping strategies to climate variability

	Structure impact of climate change	Create incentive framework	Provide access to external resources
Mariru women self-help group	Diversification of income sources (timber, goats, milk, crafts, Roscas, savings) and reduction of soil erosion	Communal pooling of knowledge, seeds, and access to local market	Group application to a community development grant to dig a dam
Kamuiritiri self-help group	Creates, maintains, and manages access to the local dam; diversification of income sources (Roscas and savings)	Frames water as a common pool resource	Obtained financial and technical help to build and maintain the dam
Kamurugu project	Diversification of income sources (grafted mangoes and papaya, honey, goats, poultry), provides education regarding agriculture	Communal pooling of agricultural products and knowledge, seed, develop new model for collective action, and access to local and regional market	Train farmers to apply to community development grants and to access non-local markets

activities, the RPO. RPO members pool resources to access regional markets and agricultural knowledge.

Fewer RPOs develop adaptation practices of storage, exchange, or mobility. The Kamuirutiri self-help group, for example, is one of the very few RPOs promoting the adaptation practice of storage as a direct response to climate variability and change. This RPO has developed a communal dam to store water. The relative old age of this RPO, created in the 1990s, may explain this difference. The microfinance model that promotes income diversification had not reached the Mt. Kenya area at the time of its creation. Similarly, few groups organize exchange activities except for marginal exchanges of tools between farmers. Work on social network analysis and coping strategies, not detailed here, shows that the functions of storage and exchange constitute traditional coping strategies which are undertaken by institutions or organizations other than RPOs, such as traditional mutual help networks which are based on kin or family ties (Washington-Ottombre 2010). A few groups promote mobility by organizing transport of goods to local markets.

Interview data on pathways for future adaptation suggest that RPOs are currently prevalent in shaping drought coping strategies and may continue to fill that role. The case of the Kamurugu project, a large RPO, is particularly enlightening in that respect. In response to the 1984 drought that devastated the area, the local Catholic mission initiated this RPO based near Gatumbiri. Since 2000, the Kamurugu project has emancipated itself from the supervision of the church and operates without the help of any external donor. The initial goal of the Kamurugu project in the early 1990s was to train farmers on how to grow drought-resistant cash crops. Subsequently, it has incorporated the training in dairy goat rearing, poultry keeping, and honey harvesting and commercializing. The Kamurugu project combines the activities of an agricultural extension organization, a community social service, and an agricultural cooperative. It plays a crucial role in crafting current and future vulnerability and adaptation to climate change. First, it structures the impacts of climate risks on households by encouraging the cultivation of drought-resistant crops and developing new income-generating activities. Second, and most predominantly, it has shaped responses to climate change in an original, collective manner. The Kamurugu project has played a major role in institution building by teaching people how to organize themselves in autonomous groups to maintain the benefits of the initial training. Farmers have then diversified the activities of the original group and/or adopted new forms of collective action for other activities. Finally, it mediates the influence of external interventions on adaptation practices by providing support in grant writing and interface between the farmers, governmental agricultural extension services, and external donors. By doing so, the Kamurugu project crafted a number of adaptation measures, such as storage practices, by encouraging the drying of fruits and vegetables, the diversification of income-generating activities (e.g. fruits, goat milk, honey…), and the communal pooling of agricultural knowledge, seeds, and seedlings.

Role-playing games

The RPGs reveal that RPOs greatly affect local strategies for coping with climate variability and may deeply influence future adaptation to climate change (Table 2). Farmer's strategies observed during the game show that RPOs are key to farmers' decisions and may help them to develop non-climate-dependent activities in a future context of climate change. In non-drought seasons, RPOs help farmers build financial assets. During prolonged droughts, the function of RPOs switches toward new and less climate-dependent income-generating activities.

During the first four seasons of the game, when a regular bimodal climate was simulated, RPOs played a more important role in semi-arid areas than in humid areas. In both Ndunduri and Kambita villages, game participants

Table 2 Timeline of the role-playing games on adaptation to climate change in Kambita and Ndunduri villages

Season	Weather conditions	Kambita (semi-arid)	Ndunduri (humid)
1	Normal rains	Normal farming practices. After the rains, creation of RPOs (self-help and saving groups) to open small businesses and a collective bank account	Normal farming practices. After the harvest, people place their money in bank accounts and keep some of their crops in their granary
2	Normal rains	Same as previous season	Same as previous season
3	Normal rains	The RPO extends its activity to develop a small-scale irrigation project. Economic well-being improves	Same as previous season
4	Normal rains	Same as previous season	Same as previous season
5	Drought	After the drought, people consume their financial and physical capital (food, money, and livestock). People borrow money from family members, groups and/or the bank	After the drought, people consume their financial and physical capital (food, money, and livestock). Those who are irrigating are still able to harvest
6	Drought	Continue to consume their stock and start to become destitute. People look for casual jobs and migrate in search of employment	Continue to consume their stock and withdraw money from the bank. Some borrow money from RPOs (saving groups). People start new RPOs (ROSCAs)
7	Drought	Creation of RPOs (self-help group) to plant earlier and dig deeper using collective labor, to weave baskets and ropes, and to burn charcoal. People continue to consume their stock and become more and more destitute. They look for food in the forest and kill antelopes. Some borrow seeds from friends	Continue to consume their stock, withdraw money from the bank, and borrow money from RPOS. People look for casual jobs and migrate in search of employment. Some start to become destitute
8	Drought	Creation of a RPO (self-help group) to conserve water for irrigation but failure due to lack of capital and drought. Some decide to create a RPO (self-help group) to cultivate aloe vera (a drought-resistant crop)	Creation of RPOS to rear chicken, farm fish, and weaving baskets. People start new RPOs (ROSCAs). People continue to migrate to urban centers in search for employment
9	Drought	Because of the cultivation of aloe vera, people start paying their debts back and put children back to school	Same as previous season. Creation of new RPO (self-help) to grow vegetables using domestic water for irrigation. Economic well-being improves
10	Drought	Same as previous season. Economic well-being improves. End of the simulation	Same as previous season. Creation of RPOs (self-help) to dig a dam and to sell drought-resistant crops
11	Drought	N/A	Same as previous season. Some people uproot their tea plants to create space for drought-resistant crops. Creation of a large RPO (self-help) to develop irrigation
12	Drought	N/A	Same as previous season. End of the simulation

tended to take advantage of a favorable climate and a good harvest by increasing their assets and stock. In Ndunduri village (humid area), where most farmers are integrated into a formal banking system due to the cultivation of tea and coffee, farmers chose to put their money in their bank account. In contrast, in Kambita village (semi-arid area), farmers created or used existing RPOs to invest the income from the good harvests into the creation of new income-generating activities. For instance, a group of game participants decided right after the first harvest to form a saving group and ROSCAs in order to create small businesses to provide additional income in case of droughts. After the third good season, they created a new RPO to sell maize at the market and place the money in a collective bank account. Another group of game participants in Kambita village decided after three good seasons to create a self-help group to plant sunflowers and rear dairy cows. RPOs seemed to be particularly important in "normal" climatic conditions in areas where farmers do not usually have access to the formal banking system. In regular climatic conditions, RPOs participated in the adaptation practices of storage, diversification, and communal pooling.

During drought years, RPOs supported the development of non-climate-dependent income-generating activities in both the semi-arid and humid areas. For instance, in Ndunduri village, after three seasons of droughts, game participants created several self-help groups to promote activities such as rearing chickens, weaving baskets, and developing fish farms. Observing that farmers using irrigation were not as affected by the droughts as others, game participants created a self-help group to use domestic water to irrigate vegetables for sale on the market. The following season, game participants modified the functions of this

RPO toward a village-wide irrigation and dam project to collect snow-melt water for horticulture. In Kambita village where RPOs were highly prevalent in "normal" climatic conditions, they continued to play a major role in adaptation strategies during the simulated prolonged drought. In the simulations done in Kambita village, farmers created a self-help group to cultivate *aloe vera*, a drought-resistant cash crop currently spreading on the slopes of Mt. Kenya and in other dry areas of Kenya (Watson and Binsbergen 2008). Farmers first created a group to request a loan from external donors in order to buy the seedlings. They could not have access to the loan if they had not constituted a group. The first season, they planted on a communal field, worked, harvested, and shared the profits equally among members. In each subsequent season, new seedlings were collected from the original field and planted on individual plots. After two seasons, the group also bought a machine to process the *aloe vera* directly and increases their benefit.

Household surveys

Table 3 summarizes the statistics of participation in rural organizations and the demographic variables used in our analysis. Note that households in Gatumbiri and Kambita participate much more in rural organizations than those in Ndunduri (60.8, 75.3, and 38.6 %, respectively). Most survey respondents in all villages did not receive a formal education beyond the primary level. The size of the land parcels was much larger in the dry areas than in Ndunduri. In Kambita, both heads of households were farmers and derived all of their income from farming. The situation in each of the other two villages is less uniform, probably due to the relatively large cash income created by the culture of miraa in Kambita and the absence of other income-generating activities. In all villages, the main activity of the heads of households was mostly locally based. As discussed above, farmers received information about farming and other income-generating activities most frequently from RPOs and individuals rather than from government officials, governmental agencies, or NGOs. However, villages did differ. In particular, we found that more households in Gatumbiri declared having access to information from NGOs. This may be explained by the village's geographic proximity to the Kamurugu Agricultural Development Initiatives (KADI), a major NGO in the area.

The type and level of access to the rural organizations by community vary (Table 4). Most households have access to information about farming and other income-generating activities from government officials either more than once a year but less than once a month, once a year, or never. The same was true for governmental agencies. However, households have access to information about farming and other income-generating activities from rural organizations approximately once a week in all three communities. In Gatumbiri village, 13.6 % of households never have access to information about farming and other income-generating activities from rural organizations. Most households never have access to information about farming and other income-generating activities from NGOs except households in Gatumbiri village. Finally, most households have access to information about farming and

Table 3 LRI participation and demographic information for survey respondents

Name of variable	Gatumbiri	Kambita	Ndunduri
Number of respondents	125	77	197
Participation in rural organizations			
No org	0.0 %	1.3 %	0.5 %
Church group	19.2 %	11.7 %	13.2 %
ROSCAs	20.0 %	11.7 %	10.7 %
Cooperative	0.0 %	0.0 %	37.1 %
RPOs	60.8 %	75.3 %	38.6 %
Demographics			
Gender			
Male	24.0 %	35.1 %	40.6 %
Female	76.0 %	64.9 %	59.4 %
Age*	43.40	42.55	46.25
Education			
No formal education	5.2 %	5.2 %	17.9 %
Some primary	44.2 %	44.2 %	38.3 %
Completed primary	33.8 %	33.8 %	21.4 %
Some secondary	6.5 %	6.5 %	4.1 %
Certificate level	10.4 %	10.4 %	2.0 %
Above certificate	0.0 %	0.0 %	1.5 %
Land holdings			
Size of land (in ha)*	3.6	3.4	1.3
Main occupation			
Both farmers	46.4 %	85.7 %	59.9 %
One farmer	38.4 %	13.0 %	23.9 %
Both not farmers	15.2 %	1.3 %	16.2 %
Income from farming			
All from farming	5.6 %	53.2 %	21.8 %
Most from farming	37.6 %	29.9 %	41.1 %
Half from farming	7.2 %	10.4 %	10.7 %
Most not from farming	36.8 %	6.5 %	24.4 %
None from farming	12.8 %	0.0 %	2.0 %
Main location			
Both in the village	78.4 %	77.9 %	79.2 %
One in the village	20 %	22.1 %	17.8 %
None in the village	1.6 %	0.0 %	3.0 %
Remittances*	0.488	0.325	0.239

* Indicates continuous variables where means are given as descriptive statistics

Table 4 Respondents' sources of information for farming and other income-generating activities

Name of variable	Gatumbiri (%)	Kambita (%)	Ndunduri (%)
Government officials			
Never	24.0	32.5	41.6
Less than once a year	0.8	0.0	1.0
Once a year	4.8	3.9	9.6
More than once a year	31.2	20.8	17.8
Once a month	15.2	15.6	8.6
More than once a month	19.2	16.9	10.7
Once a week	1.6	7.8	5.1
More than once a week	3.2	2.6	5.6
Governmental agencies			
Never	24.8	23.4	23.9
Less than once a year	2.4	0.0 %	0.5 %
Once a year	9.6 %	11.7 %	17.3 %
More than once a year	26.4	27.3	27.4
Once a month	11.2	14.3	13.2
More than once a month	16	11.7	10.2
Once a week	5.6	10.4	2.0
More than once a week	4.0	1.3	5.6
Local rural organizations			
Never	13.6	1.3	0.5
Less than once a year	0.0	0.0	0.0
Once a year	0.0	0.0	0.0
More than once a year	0.0	0.0	0.5
Once a month	0.8	0.0	0.5
More than once a month	0.0	0.0	0.5
Once a week	76.8	87.0	77.3
More than once a week	8.8	11.7	20.6
NGOs			
Never	25	77.9	96.4
Less than once a year	1.6	0.0	0.0
Once a year	5.6	10.4	0.5
More than once a year	28.0	2.6	1.0
Once a month	10.4	5.2	0.0
More than once a month	18.4	2.6	0.0
Once a week	7.2	0.0	1.0
More than once a week	3.2	1.3	0.0
Individuals			
Never	6.4	6.4	5.6
Less than once a year	0.0	0.0	0.0
Once a year	0.0	1.3	0.0
More than once a year	18.4	13	7.6
Once a month	5.6	3.9	7.1
More than once a month	4.8	6.5	7.6
Once a week	10.4	13	15.2
More than once a week	54.4	55.8	56.9

other income-generating activities from other individuals more than once a week in all three communities.

CART models for each of the communities are depicted in Figs. 3, 4, and 5. Each tree shows how relevant splits in the data explain the relationships between the dependent and independent variables. The most significant splits are made first and arranged at the top of the tree. Subsequently, less important splits are made afterward and arranged lower in the tree. At the top of each tree, the mean indicated in the box titled "LRI" shows the participation in rural organizations that promote LRIs in each village. Lower branches of the tree specify which variables condition participation in those organizations and are arranged from the most to the least influential. Each branch splits the data in two groups, one that participates more and one that participates less in those organizations. The mean reported in the box of each group provides an indication regarding the direction of the relationship between the dependent and the independent variables. The higher the mean, the more likely are survey respondents to participate in those organizations.

Gatumbiri village (Fig. 3) produced the largest CART model with 12 nodes. The largest separation occurred at the level of the variable "groups" with those participating once a week or more (>7) having access to a wider and more complex range of rural organizations (mean of 2.096 vs. 4.381). Age was the next best discriminator with those older than 37 years old having the greatest participation in rural organizations (mean of 4.567 vs. 4.081). For younger heads of households, those who had access to information on farming and other income-generating activities from rural organizations more than once a week (>8) participated in a wider range of rural organizations (mean of 5 vs. 3.937). Among those who had less access to information from RPOs, those who accessed information on farming and other income-generating activities from government agencies more than once a month but less than once a week (<6) participated more in rural organizations (mean of 4.12 vs. 3.286). The next separation occurs within the group of households who had less access to information about farming and other income-generating activities from local government officials. Those who had access to information on farming and other income-generating activities from a government agency more than once a year (>3) had access to a wider range of rural organizations (mean of 4.6 vs. 3.8). Finally, within the group of households who had less access to information from governmental agencies, those whose head was older than 33 years of age (but younger than 37 according to the second division) had access to a wider range of rural organizations than younger households (mean of 4.2 vs. 3).

Kambita village (Fig. 4) produced a CART model with 6 nodes. The model was best separated by households'

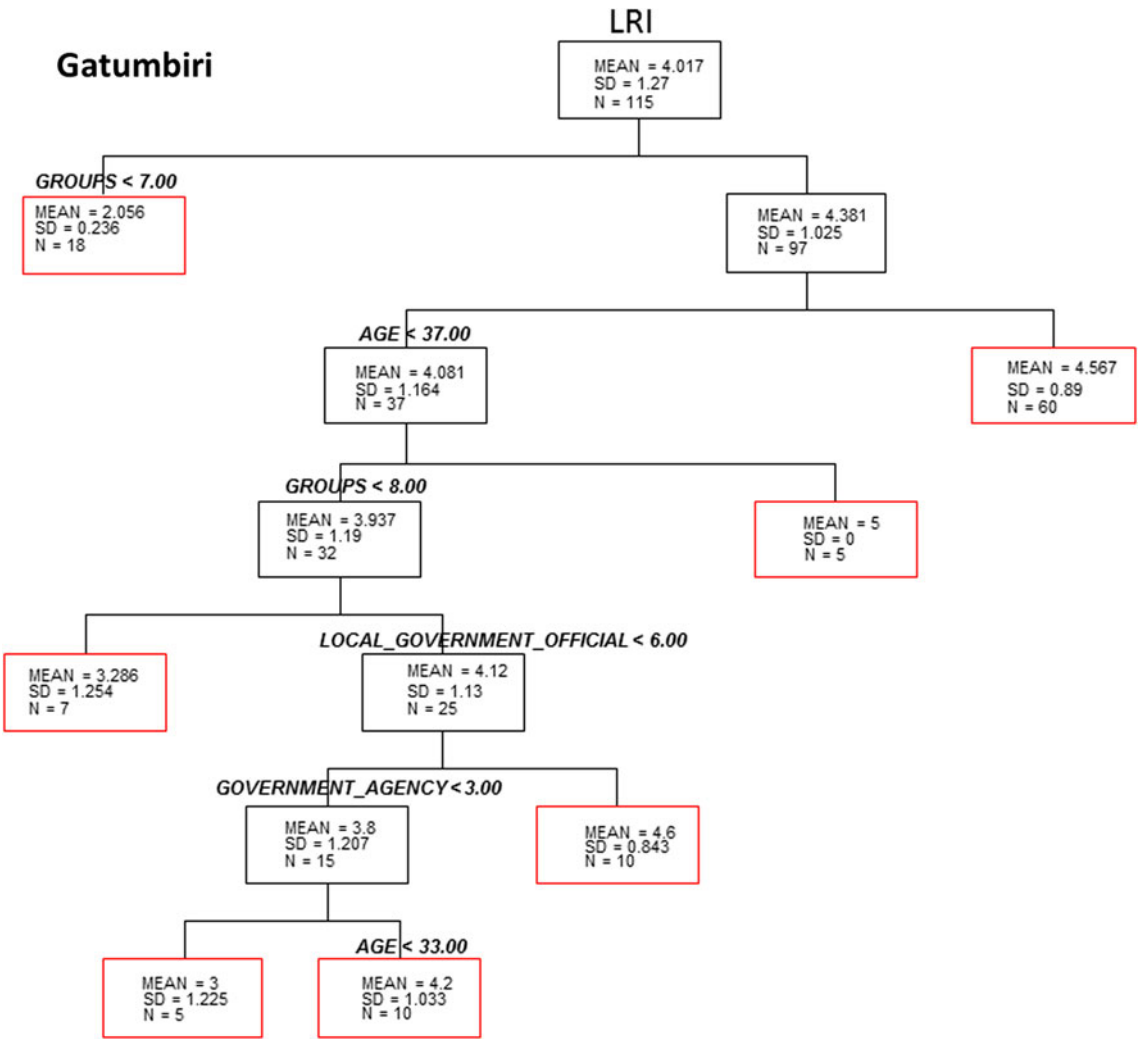


Fig. 3 Classification tree of Gatumbiri

frequency of access to information on farming and other income-generating activities from governmental agencies. Households who tended to have access to information from governmental agencies more than once a year (<3) tended to participate in a wider range of rural organizations, including RPOs (mean of 4.559 vs. 3.722). Among the households who had access to information from governmental agencies less than once a year, those who had access to information on farming and other income-generating activities from local government official more than once a year but less than once a month (<4) tended to participate in more rural organizations (mean of 4.571 vs. 3.182). On the other hand, among the households who had access to information from governmental agencies more than once a year, individuals younger than 25 years of age tended to participate less in rural organizations than others (mean of 4.63 vs. mean of 3.8).

Ndunduri village (Fig. 5) produced a CART model with 6 nodes. The largest separation occurred above and below 43 years of age. Individuals older than 43 years of age participated more in rural organizations (mean of 4.268 vs. 3.747). However, the second separation shows that individuals older than 68 years of age participated less in rural organizations (mean of 4.424 vs. 3.935). Within the age group above 68 years of age, individuals who received information about farming and other income-generating activities from governmental agencies more than once a year (>3) participated more in rural organizations than others (mean of 4.208 vs. 3).

Discussion

In this paper, we asked how LRIs promoted by rural organizations shape coping strategies to climate variability and may structure future adaptations to climate change. We investigated what types of households participate in those organizations and the consequences for their vulnerability to

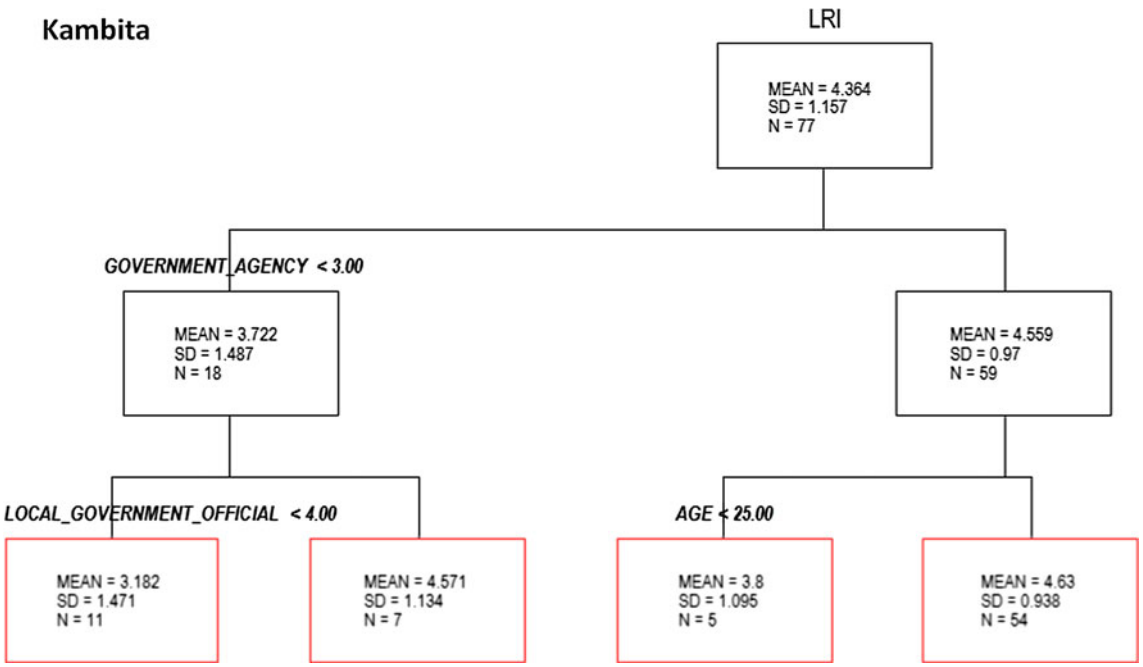


Fig. 4 Classification tree for Kambita

climate change and variability. Regarding our first research question, semi-structured interviews and role-playing games provided information on the importance as well as the dynamism of rural organizations in general and RPOs in particular in adaptation to climate change and variability. A number of short-term coping strategies and most long-term adaptation strategies relied on RPOs to develop non-climate-dependent income-generating activities.

Regarding our second research question, our statistical analysis provided a number of results demonstrating that more vulnerable households participated less in rural organizations, including RPOs. Results showed that people with less education have less access to RPOs than people with more education. It clearly appears that very young or very old households tend to participate less in RPOs than middle-aged households. The least educated and both the youth and the elderly do not participate in RPOs and may therefore be more vulnerable to climate change because they will have less opportunities to diversify their coping strategies, to pool resources with others, and to develop non-climate-dependent income-generating activities. Contrary to what most studies on gender and collective action show (e.g. Agrawal and Gupta 2005), the CART models did not show a significant effect of gender on participation in RPOs. The analysis of some of the results of the CARTS model also proved contradictory to expectations.

We addressed whether households with a more diversified and non-locally based lifestyle participated more in rural organizations. Results showed that households whose main occupation is farming tend to participate more in RPOs than households in which one or two members have a job not related to farming activities. Whereas past studies have emphasized the importance of seasonal or permanent migration (Adger 1999) and diversification toward non-farming income-generating activities (Leary 2006; Adger et al. 2006) in order to reduce households' vulnerabilities to climate change, these results bring a new perspective to the scope of diversification strategies available to rural households in order to cope with climate change. Participation in rural organizations provides new opportunities of income diversification and a greater guarantee of livelihood security to the households that are firmly grounded in rural areas. Those organizations promote new LRIs and revive local collective action. They greatly participate in the revitalization of rural areas, reduce local vulnerabilities, and strengthen adaptation capacities.

Our analysis of the access to information regarding farming and other income-generating activities suggests that rural organizations may have filled the vacuum left by government organizations. Both access to information from government officials and government agencies were significant variables in our statistical analysis. Households who have access to information from these sources less than once a year participate less in rural organizations. This refers to households that are disconnected from the local social environment and local collective action. They will most likely be the most vulnerable to climate change. In contrast, households that received frequent information from local officials participated more in rural organizations, included RPOs. Both local officials and rural organizations are

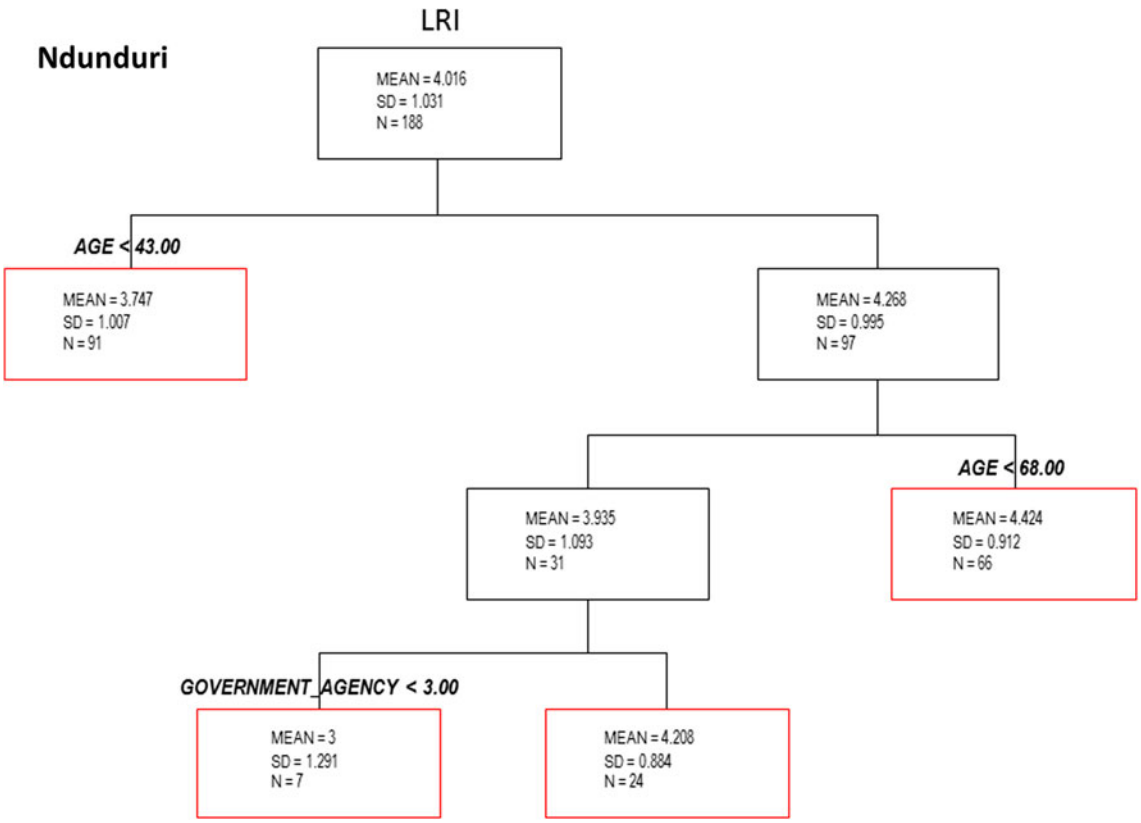


Fig. 5 Classification tree for Ndunduri

combined into a complex polycentric web of local collective action that aims at reducing vulnerabilities and enhancing adaptive capacity. Local officials (chiefs, sub-chiefs, and sub-areas chiefs) assume roles beyond their administrative functions of local leaders and promoters of livelihood security to promote income diversification and the development of non-climate-dependent income-generating activities. Local officials both rely on and encourage the development of rural organizations to support new forms of income-generating activities and livelihood security.

Finally, our results give weight to the conclusion that the participation in rural organizations may differ according to distinct institutional environments that are the result of distinct climatic and/or landscape characteristics (Agrawal 2008). The CART model of each village was strictly different. Households in humid areas seem to rely more on traditional rural organizations such as cooperatives rather than on RPOs. Their coping strategies are structured around other institutions than RPOs such as the formal banking system. The distinct organizational structure in Ndunduri village has been facilitated since the 1950s by a climatic and landscape environment that allows for the development of cash crops and other income-generating activities. In the event of climate change in the Kenyan highlands, Ndunduri farmers may not be able to rely on those organizations anymore because they may lose their traditional sources of income. In that situation, they might in fact be more vulnerable to climate change than households in semi-arid areas who are relying on newer and more adaptive rural organizations such as RPOs. It seems that through their long-standing response to climate vulnerability, both Gatumbiri and Kambita villages have demonstrated their capacity for social learning and have developed highly resilient and dynamic local collective action.

Conclusions

This study sheds light on the importance of LRIs promoted by rural organizations in shaping current responses to climate variability as well as in crafting future adaptations to climate change. The analysis of interviews, role-playing games, and household surveys contributes to the existing literature on the role of rural organizations in past, present, and future adaptation to climate change and variability. This research has demonstrated that local rural organizations in general and RPOs in particular increase livelihood security in places rendered vulnerable to climate change by an arid climate, the disengagement of governmental services, and a limited access to income-generating activities. In contrast to

common diversification and livelihood security strategies which rely on the access to urban or peri-urban structures, rural organizations allow for rural and grassroots sustainable adaptation strategies.

Our work contributes to the body of the literature on collective action and adaptation to climate change. We show that local organizations in general and RPOs in particular fill a gap left by governmental institutions and provide collective answers to climate variability and change. We argue that the value of this work is to have established the role of this new and important type of collective action in rural Kenya that significantly mitigates local vulnerabilities. We started to discover the complex polycentric institutional order that supports collective actions in rural Kenya. We found that local officials and rural organizations interact in a complex polycentric web of local collective action that aims at reducing vulnerabilities and enhancing adaptive capacity. In future work, we intend to further investigate polycentricity in rural Kenya by jointly analyzing the role of rural organizations and social networks in framing local adaptive capacity. Doing so will allow us to provide a more complete analysis of the system's resilience and adaptive capacity and could help frame more case-specific policies to enhance local adaptive capacity.

As of now, several policy implications and guidelines for development and livelihood security projects can be drawn from the present analysis. Our results confirm past studies claiming that future action should rely on local rural organizations to promote adaptation in order to benefit from synergies between projects on disaster risk management, development aid, and the reduction of inequalities. In addition to existing virtuous synergies among various initiatives, investing in successful local rural organizations constitutes a way around endemic governmental corruption, path dependency, and inefficiency. Rural organizations and RPOs are not as path dependent as governmental institutions and tend to collapse if they become dysfunctional. In that respect, local rural organizations constitute a resilient and mostly untapped resource for visibly strengthening livelihood security and adaptive capacities in rural Kenya.

References

Adams A, Cekan J, Sauerborn R (1998) Towards a conceptual framework of household coping: reflections from rural West Africa. Africa 68(2):263–283

Adger WN (1999) Social vulnerability to climate change and extremes in coastal Vietnam. World Dev 27(2):249–269

Adger WN (2006) Vulnerability. Global Environ Change 16(3): 268–281

Adger WN, Arnell NW, Tompkins EL (2005) Successful adaptation to climate change across scales. Global Environ Change 15: 77–86

Adger WN, Agrawala S, Mirza MMQ, Conde C, O'Brien K, Pulhin J, Pulwarty R, Smit B, Takahashi K (2006) Assessment of adaptation practices, options, constraints and capacity. In: Parry ML, Canziani OF, Palutikof JP, van der Linden PJ, Hanson CE (eds) Climate change 2007: impacts, adaptation and vulnerability. Contribution of working group II to the fourth assessment report of the intergovernmental panel on climate change. Cambridge University Press, Cambridge, pp 717–743

Agrawal A (2008) The role of local institutions in adaptation to climate change. IFRI working paper # W081-3

Agrawal A, Gupta K (2005) Decentralization and participation: the governance of common pool resources in Nepal's Terai. World Dev 33(7):1101–1114

Amnesty International (2007) Amnesty international report Kenya. 23 May 2007. Available at: http://www.unhcr.org/refworld/docid/46558ed120.html. Accessed 13 July 2009

Armendáriz B, Morduch J (2007) The economics of microfinance. MIT Press, Cambridge

Beckerleg S (2006) What harm? Kenyan and Ugandan perspectives on Khat. Afr Aff 105(419) 219–241

Berkes F, Jolly D (2001) Adapting to climate change: social-ecological resilience in a Canadian western arctic community. Conserv Ecol 5(2):18. [online] URL:http://www.consecol.org/vol5/iss2/art18/

Besley T, Coate S, Loury G (1993) The economics of rotating savings and credit associations. Am Econ Rev 83(4):792–810

Bosc P-M, Eychenne D, Hussein K, Losch B, Mercoiret M-R, Rondot P, Mackintosh-Walker S (2002) The role of rural producer organizations in the world bank rural development strategy. Rural development strategy background paper #8

Bouman F (1995) Rotating and accumulating savings and credit associations: a development perspective. World Dev 23:371–384

Brown M, Funk C (2008) Food security under climate change. Science 319(5863):580–581

Campbell DJ (1999) Response to drought among farmers and herders in Southern Kajiado District, Kenya: a comparison of 1972–1976 and 1994–1995. Human Ecol 27(3):377–413

Carrier N (2005) The need for speed: contrasting timeframes in the social life of Kenyan miraa. Africa 75:539–558

Chambers R, Conway G (1992) Sustainable rural livelihoods: practical concepts for the 21st century. IDS discussion paper 296, IDS, University of Sussex, Brighton, UK

Corbett J (1988) Famine and households coping strategies. World Dev 16(9):1099–1112

De'ath G, Fabricius KE (2000) Classification and regression trees: a powerful yet simple technique for ecological data analysis. Ecology 81(11):3178–3192

Easterling WE, Aggarwal PK, Batima P, Brander KM, Erda L, Howden S, Kirilenko A, Morton J, Soussana J-F, Schmidhuber J, Tubiello FN (2007) Food, fibre and forest products. Climate change 2007: Impacts, adaptation and vulnerability. In: Parry ML, Canziani OF, Palutikof JP, van der Linden PJ, Hanson CE (eds) Contribution of working Group II to the fourth assessment report of the intergovernmental panel on climate change. Cambridge University Press, Cambridge, pp 273–313

Eriksen S, Brown K, Kelly M (2005) The dynamics of vulnerability: locating coping strategies in Kenya and Tanzania. Geogr J 171(4):287–305

Gugerty MK (2007) You can't save alone: commitment in rotating savings and credit associations in Kenya. Econ Dev Cultural Change 55(2):251–282

Haugerud A (1993) The culture of politics in modern Kenya. Cambridge University Press, Cambridge

Holstein J, Gubrium J (2003) Inside interviewing: new lenses, new concerns. Sage Publication, CA

Kelly P, Adger N (2000) Theory and practice in assessing vulnerability to climate change and facilitating adaptation. Clim Change 47:325–352

Leary N, Adejuwon J, Bailey W, Barros V, Caffera M, Chinvanno S, Conde C, De Comarmond A, De Sherbinin A, Downing T, Eakin H, Nyong A, Opondo M, Osman B, Payet R, Pulhin F, Pulhin J, Ratnasiri J, Sanjak E, von Maltitz G, Wehbe M, Yin Y, Ziervogel G (2006) For whom the bell tolls: vulnerability in a changing climate. A Synthesis from the AIACC project. AIACC working paper no. 21. Florida: International START Secretariat

Maitima JM, Mbugua SM, Atieno F, Mathai S (2004) Impacts of land use on vegetation species composition, distribution, structure and diversity: the case of Embu and Mbeere District, Kenya. http://www.lucideastafrica.org/publications/Maitima_LUCID_WP32.pdf

Mimuyu PK (1999) Rotating saving and credit associations in rural east Africa. World Dev 7:1299–1308

Moore N, Alagarswamy G, Pijanowski B, Thornton P, Olson J, Lofgren B, Andresen J, Campbell D (2010) Food production risks associated with land use change and climate change in East Africa. Earth Environ Sci 6. IOP Publishing Ltd. doi:10.1088/1755-1307/6/4/342003

Moris J (1991) Extension Alternatives in Tropical Africa. Overseas Development Institute, London

Mwaniki HSK (1975) The living history of the Embu and Mbeere people. East Afr Lit Bureau

Olson JM, Butt B, Atieno F, Maitima J, Smucker TA, Muchugu E, Murimi G, Hong X (2003) Multi-scale analysis of the root causes of land use and cover change in Embu and Mbeere Districts, Kenya. LUCID working paper series number 20. http://www.lucideastafrica.org/publications/Olson_LUCID_WP20.pdf

Olson J, Alagarswamy G, Andresen J, Campbell D, Davis A, Ge J, Huebner M, Lofgren B, Lusch D, Moore N, Pijanowski B, Qi J, Thornton P, Torbick N, Wang J (2008) Integrating diverse methods to understand climate-land interactions in East Africa. GeoForum 39:898–911

Osman Elasha B (2006) Human resilience to climate change: lessons for Eastern and Northern Africa. A final report of AIACC project no. AF 14. International START Secretariat, Washington, USA

Ostrom Elinor (2005) Understanding institutional diversity. Princeton University Press, Princeton

Ostrom E (2010) A long polycentric journey. Ann Rev Polit Sci 13(1):1–23

Schneider SH, Semenov S, Patwardhan A, Burton I, Magadya CHD, Oppenheimer M, Pittock AB, Rahman A, Smith JB, Suarez A, Yamin F (2007) Assessing key vulnerability and the risk from climate change" in intergovernmental panel on climate change (IPCC), (2007) In Parry ML, Canziani OF, Palutikof JP, van der Linden PJ, Hanson CE (eds) Climate change 2007: impacts, adaptation and vulnerability. Contribution of working group II to the fourth assessment report. Cambridge University Press, Cambridge

Shaftel F, Shaftel G (1967) Role-Playing for the social values: decision-making in the social studies. Prentice-Hall, NJ, p 431

Smithers J, Smit B (1997) Human adaptation to climatic variability and change. Global Environ Change 7(2):129–146

Uphoff N, Buck L (2006) Strengthening rural local institutional capacities for sustainable livelihoods and equitable development. Paper prepared for the Social Development Department of the World Bank

Washington-Ottombre C (2010) The role of local institutions in shaping household responses to climate change and variability: a case study along the slopes of Mt. Kenya. Ph.D. Thesis, Purdue University

Washington-Ottombre C, Pijanowski B, Campbell D, Olson J, Maitima J, Musili A, Kibaki T, Kaburu H, Hayombe P, Owango E, Irigia B, Gichere S, Mwangi A (2010) Using a role-playing game to inform the development of land-use models for the study of a complex socio-ecological system. Agricult Syst 103(3):117–126. doi:10.1016/j.agsy.2009.10.002

Watson DJ, van Binsbergen J (2008) Livelihood diversification opportunities for pastoralists in Turkana, Kenya. ILRI Research Report 5. ILRI (International Livestock Research Institute), Nairobi, Kenya, p 43